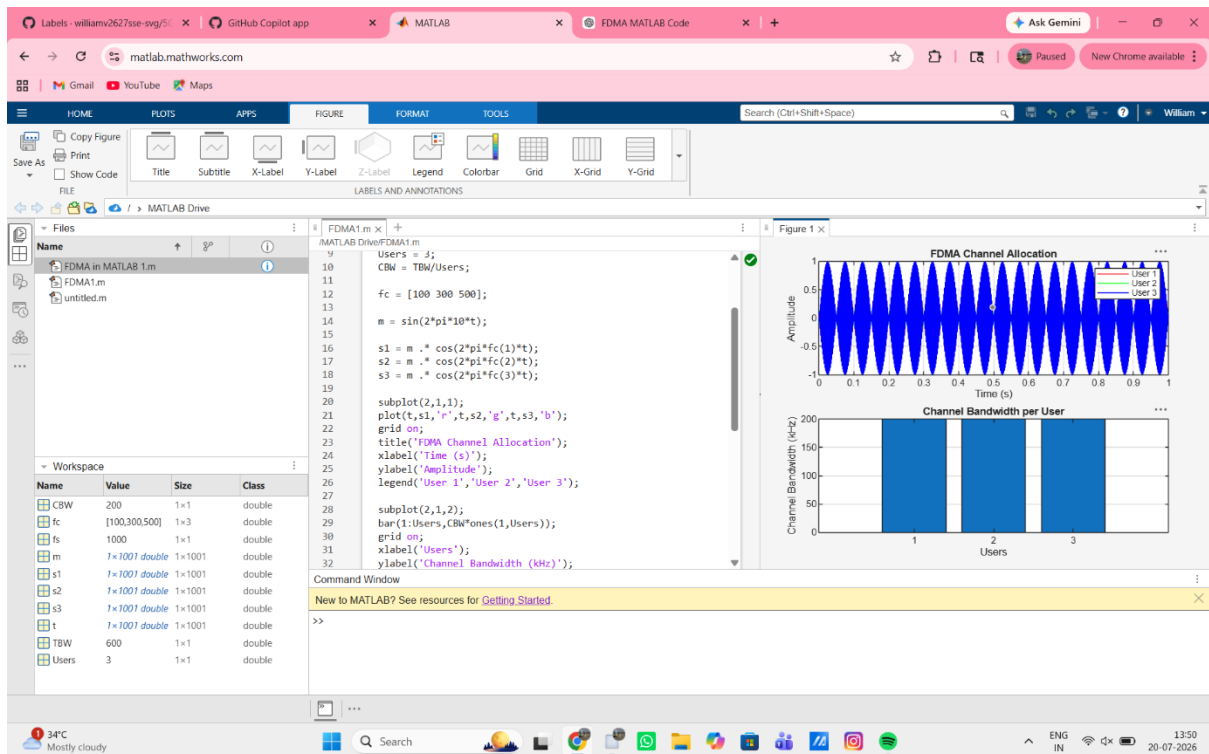


EXP.NO:01 Simulation of FDMA in MATLAB Date

(A)PROGRAM:

```
clc;
clear;
close all;
fs = 1000;
t = 0:1/fs:1;
TBW = 600;
Users = 3;
CBW = TBW/Users;
fc = [100 300 500];
m = sin(2*pi*10*t);
s1 = m .* cos(2*pi*fc(1)*t);
s2 = m .* cos(2*pi*fc(2)*t);
s3 = m .* cos(2*pi*fc(3)*t);
subplot(2,1,1);
plot(t,s1,'r',t,s2,'g',t,s3,'b');
grid on;
title('FDMA Channel Allocation');
xlabel('Time (s)');
ylabel('Amplitude');
legend('User 1','User 2','User 3');
subplot(2,1,2);
bar(1:Users,CBW*ones(1,Users));
grid on;
xlabel('Users');
ylabel('Channel Bandwidth (kHz)');
title('Channel Bandwidth per User');
```

OUTPUT



(B)

clc;

clear;

close all;

BW = 200; % Channel Bandwidth (kHz)

Users = 3;

M = 2; % BPSK Modulation

% Data Rate Calculation

DataRate = BW * log2(M); % Data Rate (kbps)

% Data rate for each user

Rate = DataRate * ones(1, Users);

% Display results

disp('User Data Rate (kbps)');

disp([(1:Users)' Rate']);

% Plot

figure;

bar(1:Users, Rate);

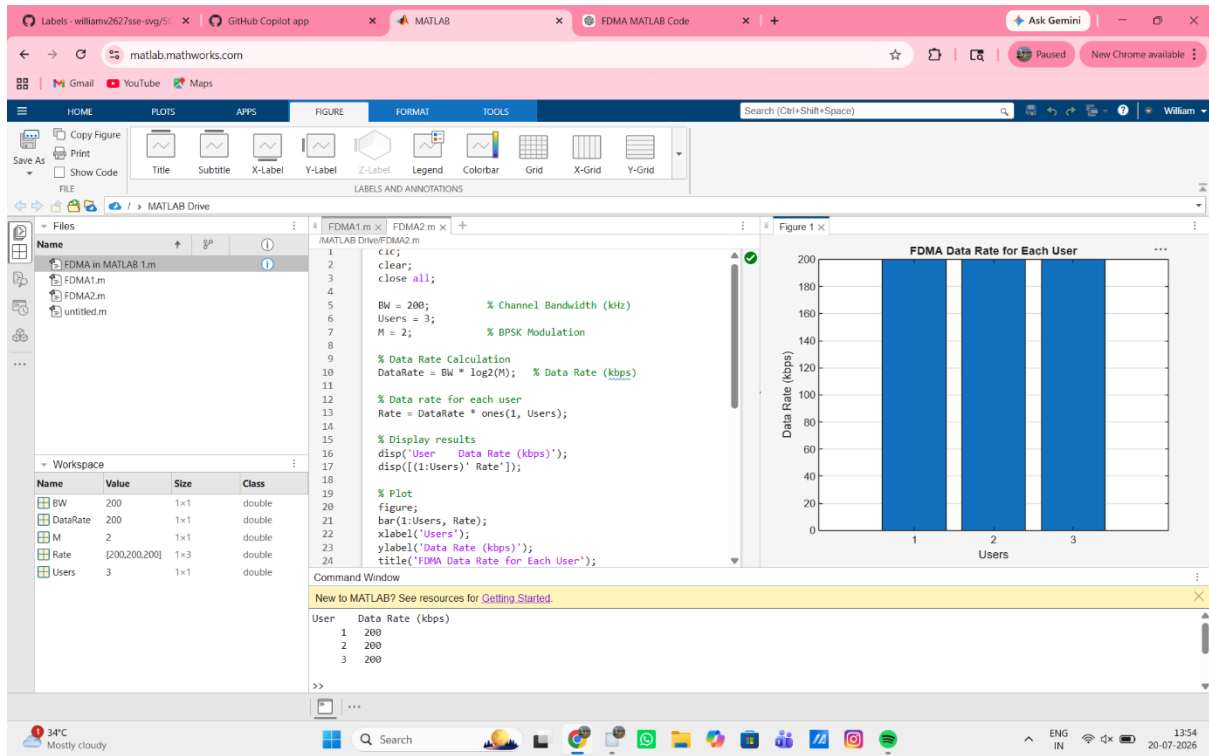
xlabel('Users');

ylabel('Data Rate (kbps)');

title('FDMA Data Rate for Each User');

grid on;

OUTPUT:



(C)

clc;

clear;

close all;

Ps = 1; % Signal Power (W)

Pn = [0.1 0.05 0.01]; % Noise Power (W)

Users = 1:3;

% Calculate SNR in dB

SNR = 10 * log10(Ps ./ Pn);

% Display results

disp('User Noise Power (W) SNR (dB)');

disp([Users' Pn' SNR]);

% Plot

figure;

```

bar(Users, SNR);

xlabel('Users');

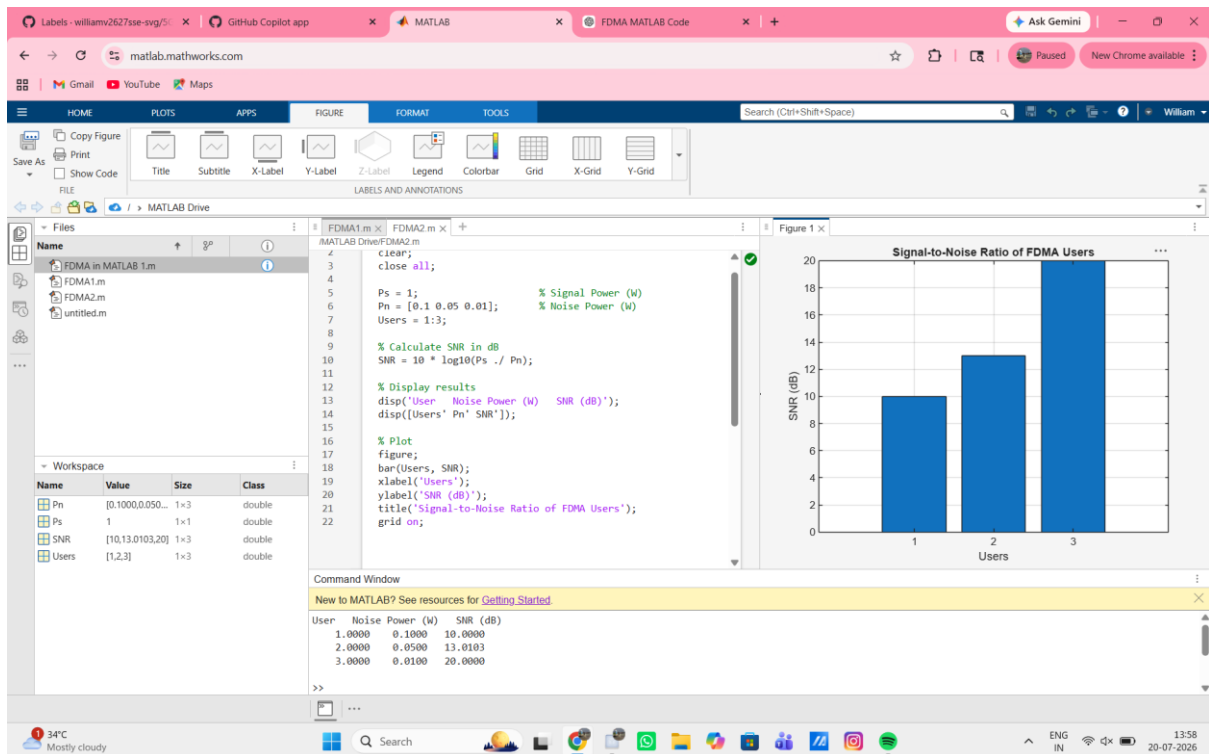
ylabel('SNR (dB)');

title('Signal-to-Noise Ratio of FDMA Users');

grid on;

OUTPUT:

```



Ex. No. 2: Implementing TDMA Slot Allocation in GSM.

Date

(A)

```

clc;
clear;
close all;

FrameTime = 4.615;    % Frame duration in ms

Slots = 8;            % Number of time slots

SlotDuration = FrameTime / Slots;

% Time vector
t = 0:SlotDuration:FrameTime;

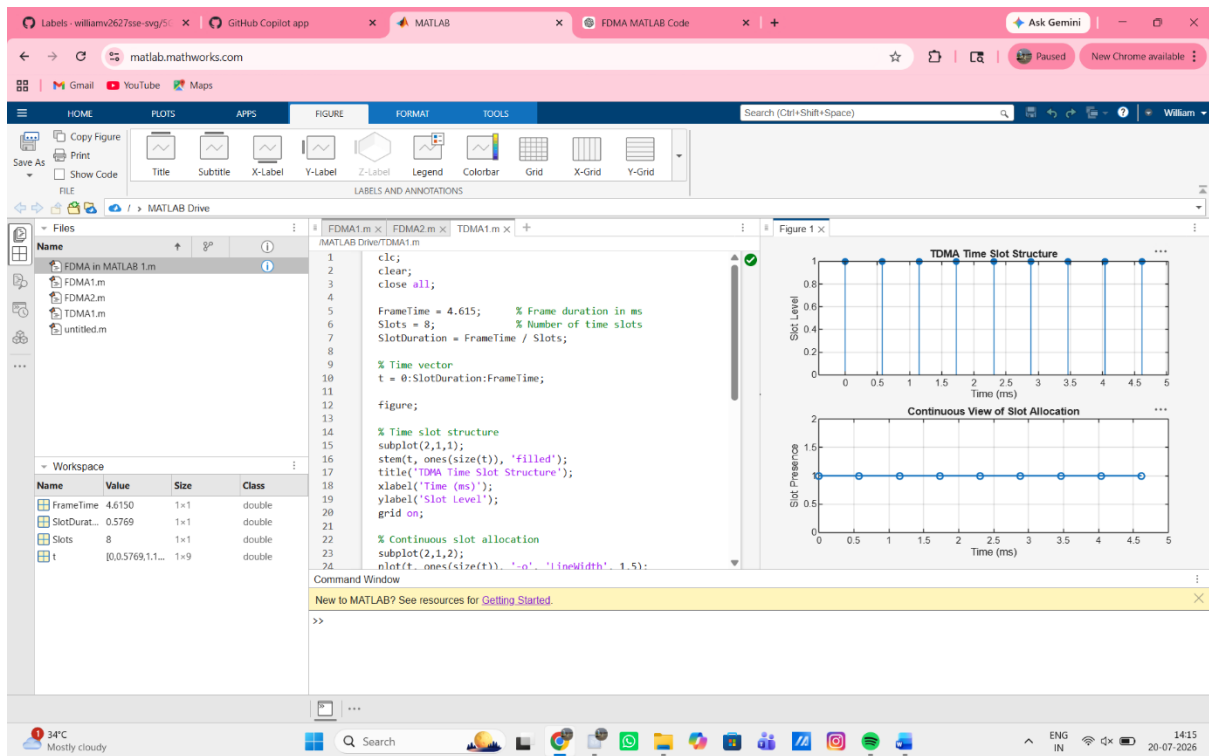
figure;

% Time slot structure
subplot(2,1,1);
stem(t, ones(size(t)), 'filled');
title('TDMA Time Slot Structure');
xlabel('Time (ms)');
ylabel('Slot Level');
grid on;

% Continuous slot allocation
subplot(2,1,2);
plot(t, ones(size(t)), '-o', 'LineWidth', 1.5);
title('Continuous View of Slot Allocation');
xlabel('Time (ms)');
ylabel('Slot Presence');
grid on;

```

OUTPUT:



(B)

clc;

clear;

close all;

TotalSlots = 8;

AllocatedSlots = 6;

FreeSlots = TotalSlots - AllocatedSlots;

% Calculate frame utilization

Utilization = (AllocatedSlots / TotalSlots) * 100;

% Display result

disp('Frame Utilization (%));

disp(Utilization);

figure;

% Graph 1: Line Plot

subplot(2,1,1);

plot([1 2], [AllocatedSlots FreeSlots], '-o', 'LineWidth', 2);

title('GSM Frame Utilization (Allocated vs Free)');

```
xlabel('Slot Type Index (1 = Allocated, 2 = Free)');
```

```
ylabel('Number of Slots');
```

```
grid on;
```

```
% Graph 2: Pie Chart
```

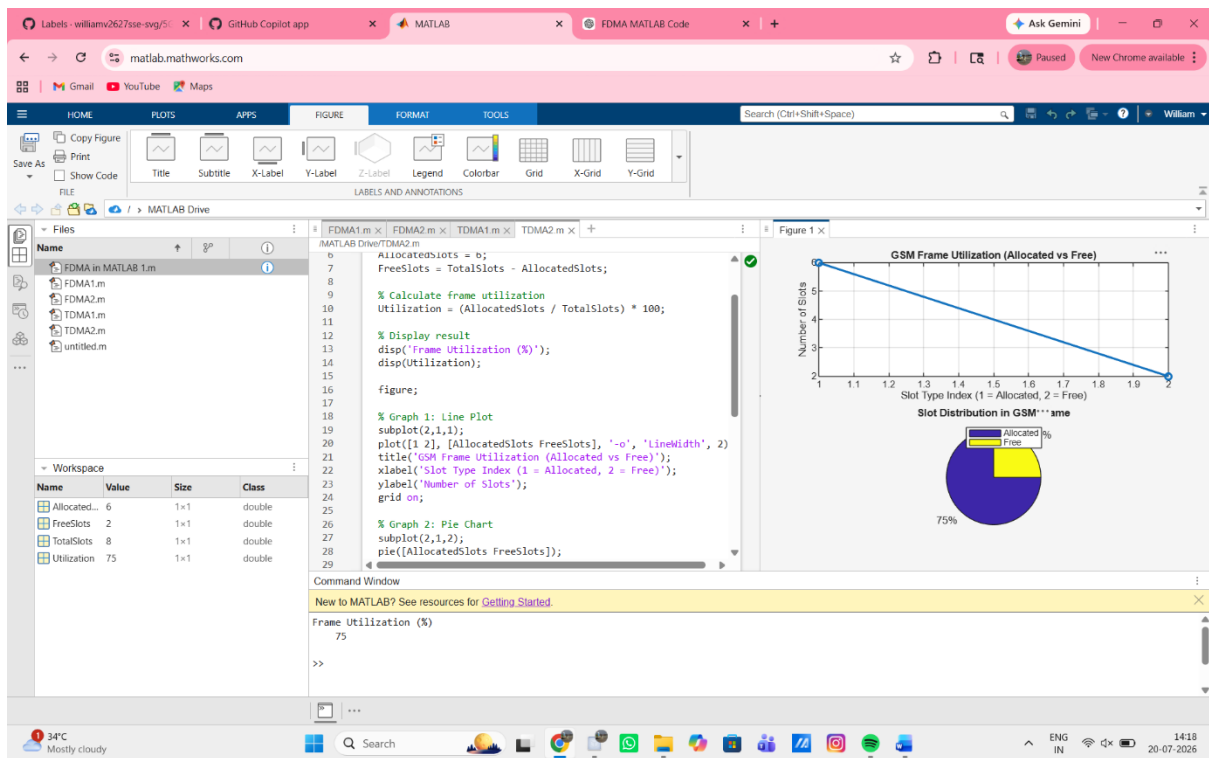
```
subplot(2,1,2);
```

```
pie([AllocatedSlots FreeSlots]);
```

```
title('Slot Distribution in GSM Frame');
```

```
legend('Allocated', 'Free');
```

```
OUTPUT:
```



(C)

```

clc;
clear;
close all;
Users = 1:8;      % Total users
TotalSlots = 5;   % Available slots
Alloc = zeros(1,8); % Initialize allocation
Alloc(1:TotalSlots) = 1; % Assign slots to first users
Blocked = 1 - Alloc; % Remaining users blocked
% Display allocation table
disp('User Wise Slot Allocation');
T = table(Users', Alloc', ...
    'VariableNames', {'User','Status_Allocated'});
disp(T);
figure;
% Graph 1: Allocation per user
subplot(2,1,1);
stem(Users, Alloc, 'filled', 'LineWidth', 2);
title('TDMA Slot Allocation per User');
xlabel('User Index');
ylabel('Slot Status (1=Allocated, 0=Blocked)');
ylim([-0.2 1.2]);
grid on;
% Graph 2: System view
subplot(2,1,2);
plot(Users, Alloc, '-o', 'LineWidth', 2);
hold on;
plot(Users, Blocked, '-s', 'LineWidth', 2);
title('TDMA Slot Utilization Analysis');
xlabel('User Index');
ylabel('Status');

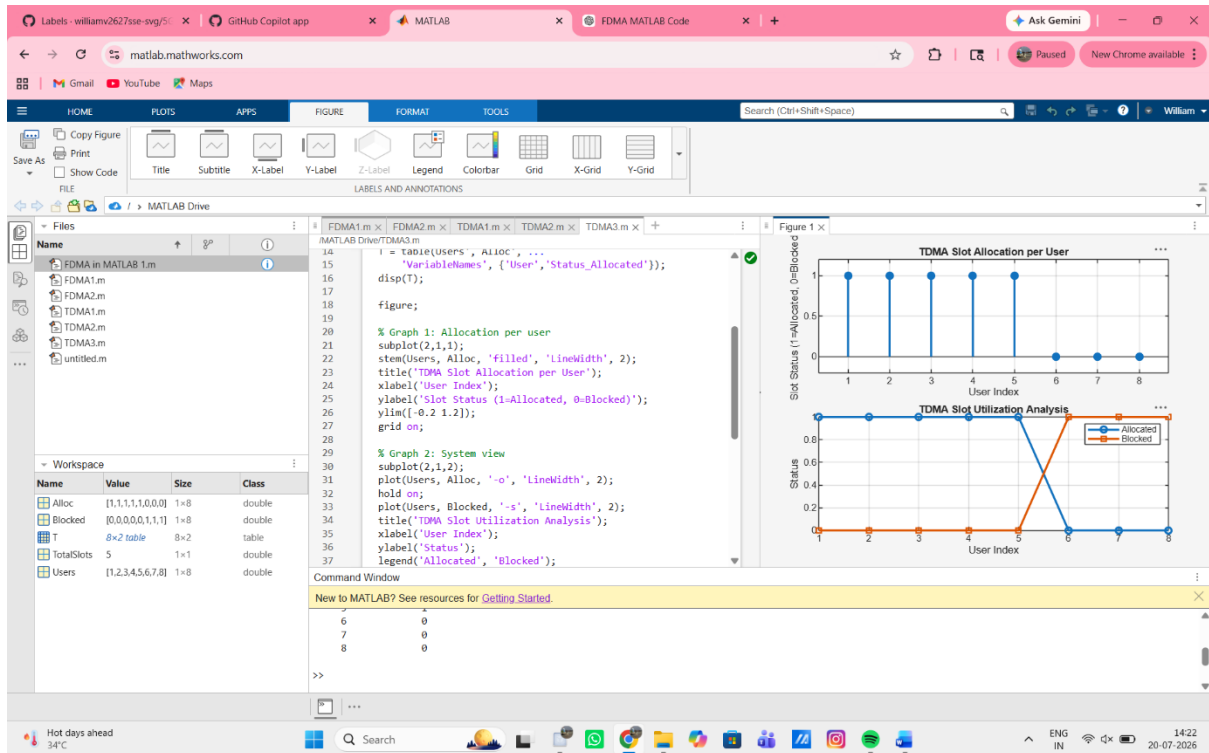
```



```
legend('Allocated', 'Blocked');
```

```
grid on;
```

OUTPUT:



Ex.N.3: Spectrum Analysis of Sub-1 GHz, Mid-band & mmWave by Plotting Propagation Loss.

(A)

```
clc;
```

```
clear;
```

```
close all;
```

```
d = 0.1:0.1:5; % Distance (km)
```

```
f = [900 3500 28000]; % Frequency (MHz)
```

```
% Free Space Path Loss (FSPL)
```

```
PL1 = 32.44 + 20*log10(f(1)) + 20*log10(d);
```

```
PL2 = 32.44 + 20*log10(f(2)) + 20*log10(d);
```

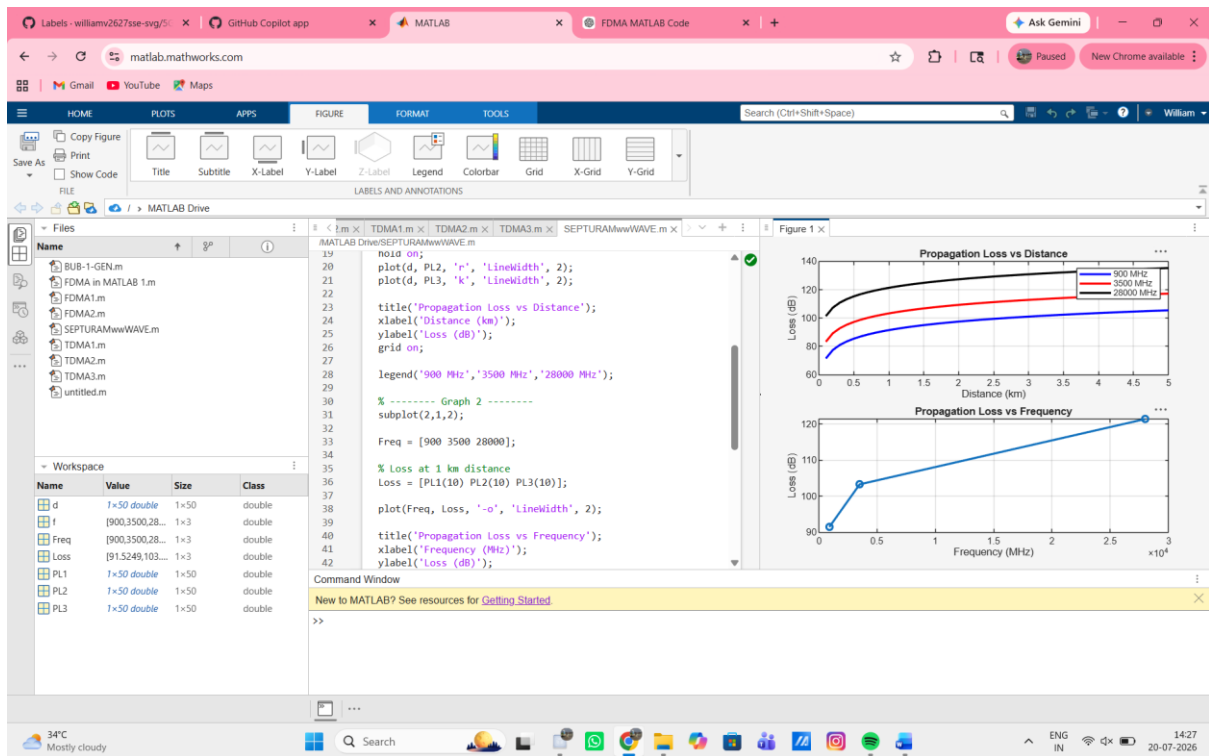
```
PL3 = 32.44 + 20*log10(f(3)) + 20*log10(d);
```

```

figure;
% ----- Graph 1 -----
subplot(2,1,1);
plot(d, PL1, 'b', 'LineWidth', 2);
hold on;
plot(d, PL2, 'r', 'LineWidth', 2);
plot(d, PL3, 'k', 'LineWidth', 2);
title('Propagation Loss vs Distance');
xlabel('Distance (km)');
ylabel('Loss (dB)');
grid on;
legend('900 MHz','3500 MHz','28000 MHz');
% ----- Graph 2 -----
subplot(2,1,2);
Freq = [900 3500 28000];
% Loss at 1 km distance
Loss = [PL1(10) PL2(10) PL3(10)];
plot(Freq, Loss, '-o', 'LineWidth', 2);
title('Propagation Loss vs Frequency');
xlabel('Frequency (MHz)');
ylabel('Loss (dB)');
grid on;

```

OUTPUT:



```
(B)

clc;

clear;

close all;

d = 2;           % Fixed distance (km)

f = [900 3500 28000]; % Frequency (MHz)

% Free Space Path Loss Calculation
PL = 32.44 + 20*log10(f) + 20*log10(d);

figure;

% ----- Graph 1: Frequency vs Loss -----
subplot(2,1,1);

plot(f, PL, '-o', 'LineWidth', 2);

title('Propagation Loss vs Operating Frequency');

xlabel('Frequency (MHz)');

ylabel('Propagation Loss (dB)');

grid on;

% ----- Graph 2: Band Comparison -----
subplot(2,1,2);

bar(PL);
```

```
title('Comparison of Propagation Loss Across Bands');
```

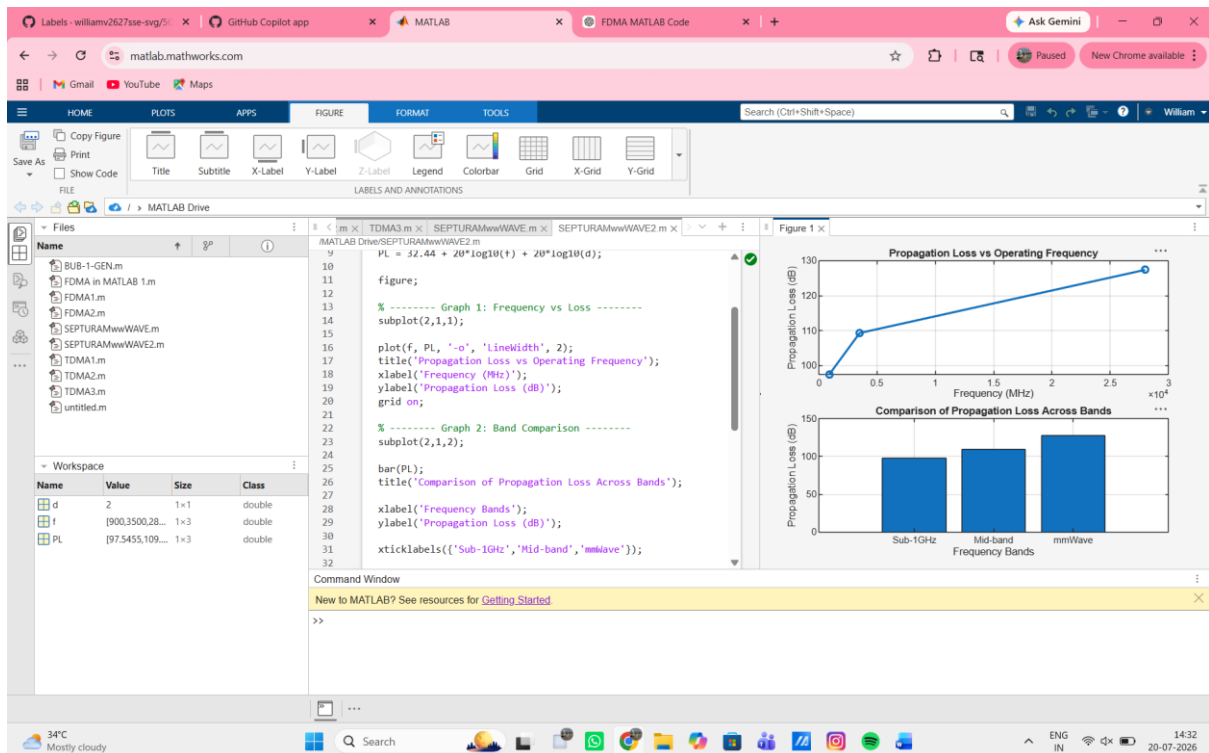
```
xlabel('Frequency Bands');
```

```
ylabel('Propagation Loss (dB)');
```

```
xticklabels({'Sub-1GHz','Mid-band','mmWave'});
```

```
grid on;
```

OUTPUT:



```
(C) clc;
```

```
clear;
```

```
close all;
```

```
% Frequency bands (MHz)
```

```
f = [900 3500 28000];
```

```
% Distance (km)
```

```
d = 2;
```

```
% Propagation Loss (FSPL)
```

```
PL = 32.44 + 20*log10(f) + 20*log10(d);
```

```
figure;
```

```
% ----- Graph 1: Frequency vs Loss -----
```

```
subplot(2,1,1);
```

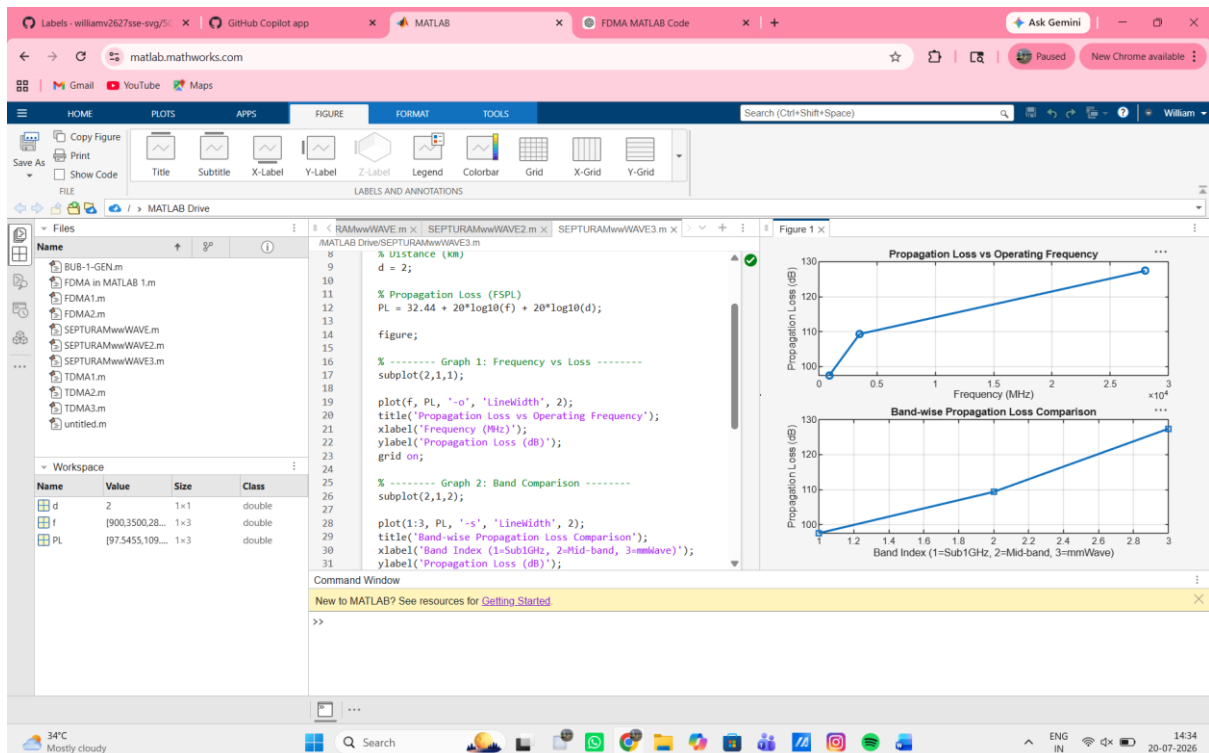
```
plot(f, PL, '-o', 'LineWidth', 2);
```

```

title('Propagation Loss vs Operating Frequency');
xlabel('Frequency (MHz)');
ylabel('Propagation Loss (dB)');
grid on;

% ----- Graph 2: Band Comparison -----
subplot(2,1,2);
plot(1:3, PL, '-s', 'LineWidth', 2);
title('Band-wise Propagation Loss Comparison');
xlabel('Band Index (1=Sub1GHz, 2=Mid-band, 3=mmWave)');
ylabel('Propagation Loss (dB)');
grid on;
OUTPUT:

```



Ex.No.4: OFDMA Resource Block Allocation Date

(A)

```
clc;
clear;
close all;
% System parameters
bandwidth = 10e6;    % 10 MHz
rb_size = 180e3;     % 180 kHz per Resource Block
% Total Resource Blocks
total_RB = floor(bandwidth / rb_size);
% Users
users = 1:6;
% RB allocation (equal + remainder distribution)
base = floor(total_RB / length(users));
alloc = base * ones(1,length(users));
remainder = mod(total_RB,length(users));
alloc(1:remainder) = alloc(1:remainder) + 1;
% RB utilization percentage
utilization = (alloc / total_RB) * 100;
% Display results
disp('Total Resource Blocks:');
disp(total_RB);
disp('RB Allocation per User:');
disp(alloc);
figure;
% ----- Graph 1: RB Allocation per User -----
subplot(2,1,1);
stem(users, alloc, 'filled', 'LineWidth', 2);
title('OFDMA Resource Block Allocation per User');
xlabel('User Index');
ylabel('Number of Resource Blocks');
grid on;
```

% ----- Graph 2: RB Utilization -----

```
subplot(2,1,2);
```

```
plot(users, utilization, '-o', 'LineWidth', 2);
```

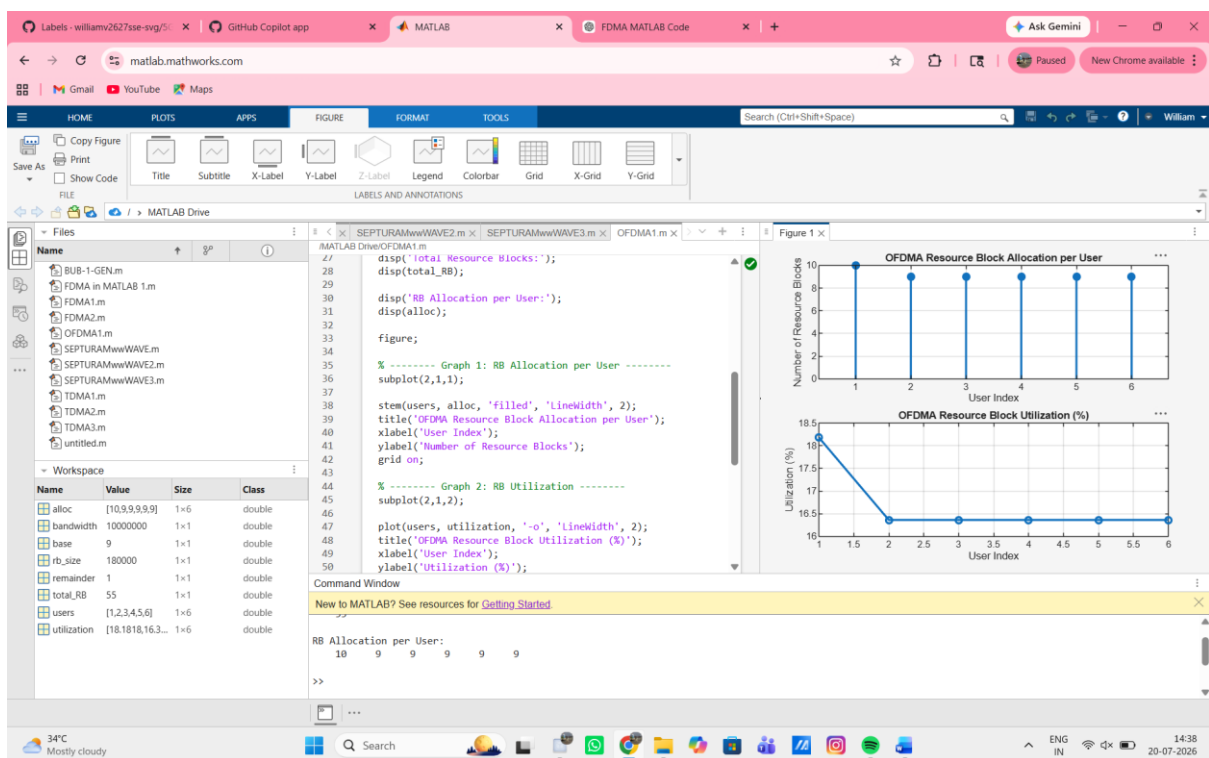
```
title('OFDMA Resource Block Utilization (%)');
```

```
xlabel('User Index');
```

```
ylabel('Utilization (%)');
```

```
grid on;
```

OUTPUT:



(B)

```
clc;
```

```
clear;
```

```
close all;
```

% Total system resource blocks

```
total_RB = 50;
```

% Number of active users

```
users = 1:5;
```

% RB allocation to users

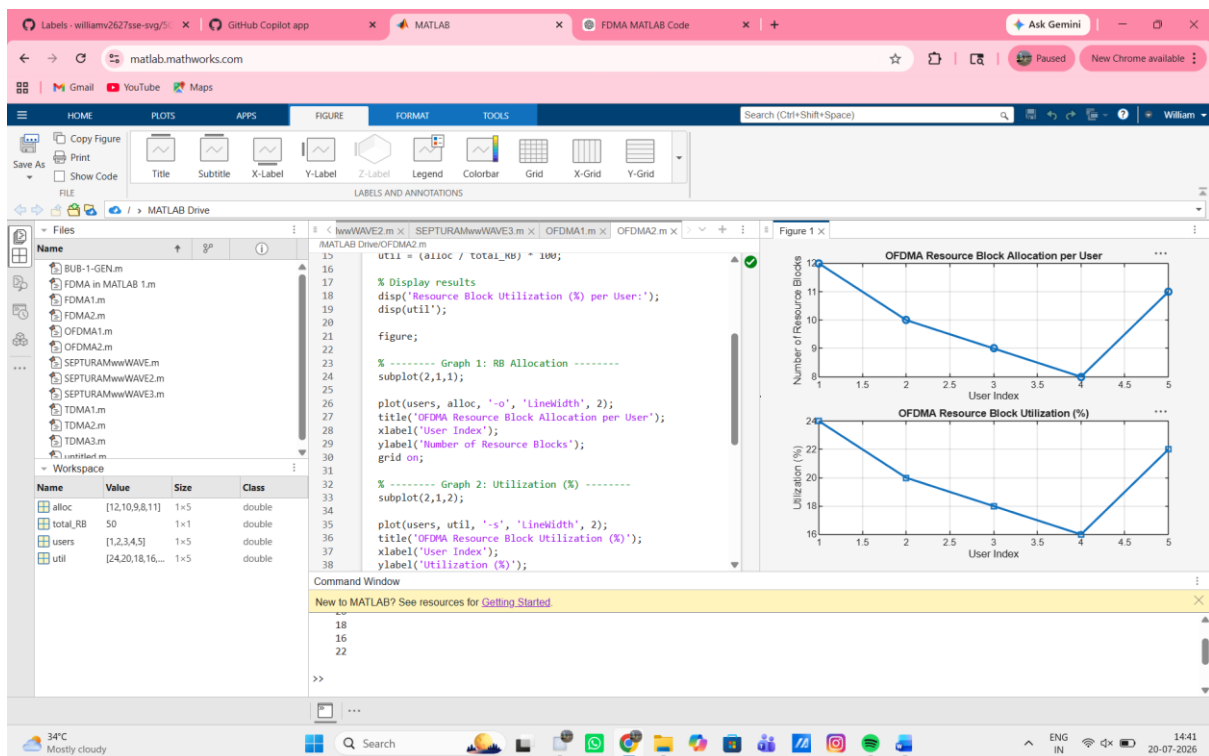
```
alloc = [12 10 9 8 11];
```

```

% Resource Block Utilization (%)
util = (alloc / total_RB) * 100;
% Display results
disp('Resource Block Utilization (%) per User:');
disp(util);
figure;
% ----- Graph 1: RB Allocation -----
subplot(2,1,1);
plot(users, alloc, '-o', 'LineWidth', 2);
title('OFDMA Resource Block Allocation per User');
xlabel('User Index');
ylabel('Number of Resource Blocks');
grid on;
% ----- Graph 2: Utilization (%) -----
subplot(2,1,2);
plot(users, util, '-s', 'LineWidth', 2);
title('OFDMA Resource Block Utilization (%)');
xlabel('User Index');
ylabel('Utilization (%)');
grid on;

```


OUTPUT:



(C)

clc;

clear;

close all;

% System parameters

rb_bw = 180e3; % Bandwidth per RB (180 kHz)

users = 1:5;

% Allocated Resource Blocks per user

alloc_RB = [12 10 9 8 11];

% Modulation efficiency (bits/symbol)

% QPSK=2, 16QAM=4, 64QAM=6

mod_eff = [2 4 6 4 2];

% Throughput calculation (Mbps)

throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;

% Display results

disp('User Throughput (Mbps):');

disp(throughput);

figure;

% ----- Graph 1: Throughput per User -----

subplot(2,1,1);

plot(users, throughput, '-o', 'LineWidth', 2);

title('OFDMA User Throughput');

xlabel('User Index');

ylabel('Throughput (Mbps)');

grid on;

% ----- Graph 2: RB vs Throughput -----

subplot(2,1,2);

plot(alloc_RB, throughput, '-s', 'LineWidth', 2);

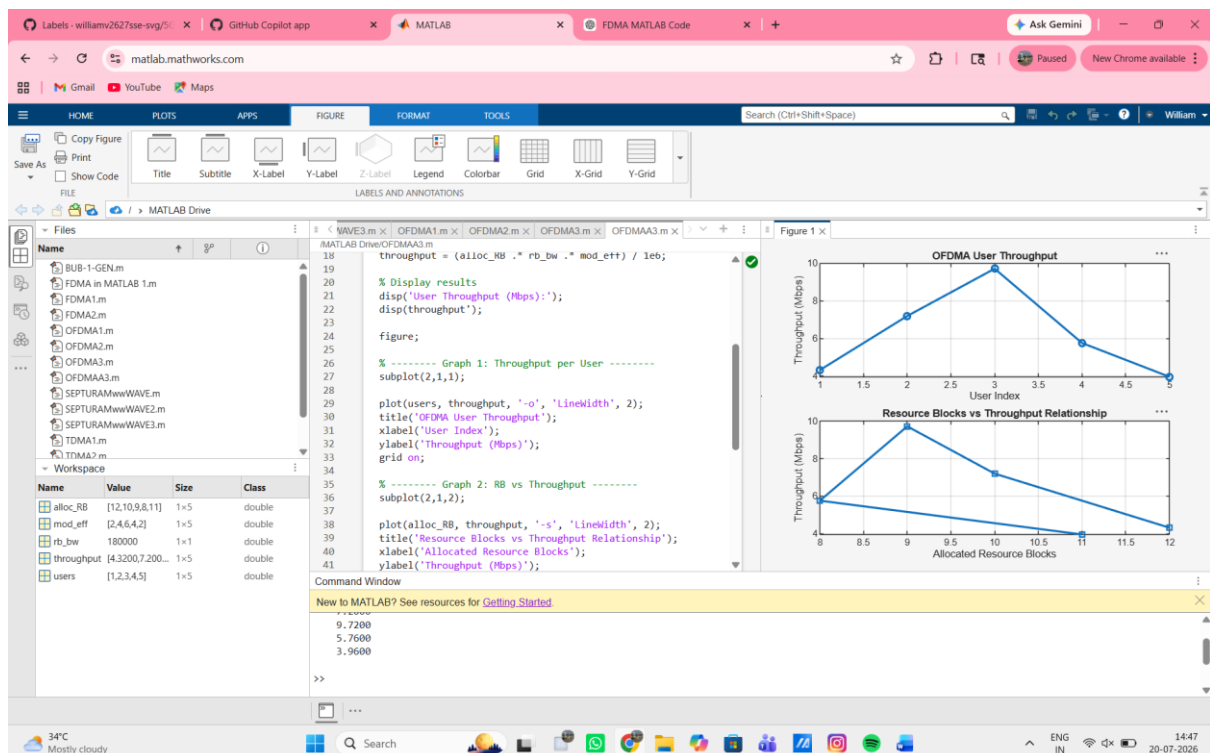
title('Resource Blocks vs Throughput Relationship');

xlabel('Allocated Resource Blocks');

ylabel('Throughput (Mbps)');

grid on;

OUTPUT:



Ex.No.5: MIMO System Simulation with BER Analysis

Date

(A)

```
clc;
clear;
close all;
% SNR range
SNR_dB = 0:2:20;
SNR = 10.^(SNR_dB/10);
% MIMO parameters
Nt = 2;
Nr = 2;
BER = zeros(1,length(SNR));
for i = 1:length(SNR)
    % Generate bits
    bits = randi([0 1],10000,1);
    % BPSK modulation
    symbols = 2*bits - 1;
    % Channel matrix
    H = (randn(Nr,Nt)+1j*randn(Nr,Nt))/sqrt(2);
    % Send same signal from both antennas
    tx = repmat(symbols',Nt,1);
    % Received signal
    rx = H*tx;
    % AWGN noise
    noise = (randn(Nr,10000)+1j*randn(Nr,10000))/sqrt(2*SNR(i));
    rx = rx + noise;
    % Combine received signals
    combined = sum(rx,1);
    % Detection
    detected_bits = real(combined) > 0;
```

```
% BER calculation
```

```
BER(i)=sum(detected_bits~=bits)/length(bits);
```

```
end
```

```
figure
```

```
% BER vs SNR
```

```
subplot(2,1,1)
```

```
semilogy(SNR_dB,BER,'-o','LineWidth',2)
```

```
title('2x2 MIMO BER Performance vs SNR')
```

```
xlabel('SNR (dB)')
```

```
ylabel('BER')
```

```
grid on
```

```
% Reliability
```

```
subplot(2,1,2)
```

```
plot(SNR_dB,1-BER,'-s','LineWidth',2)
```

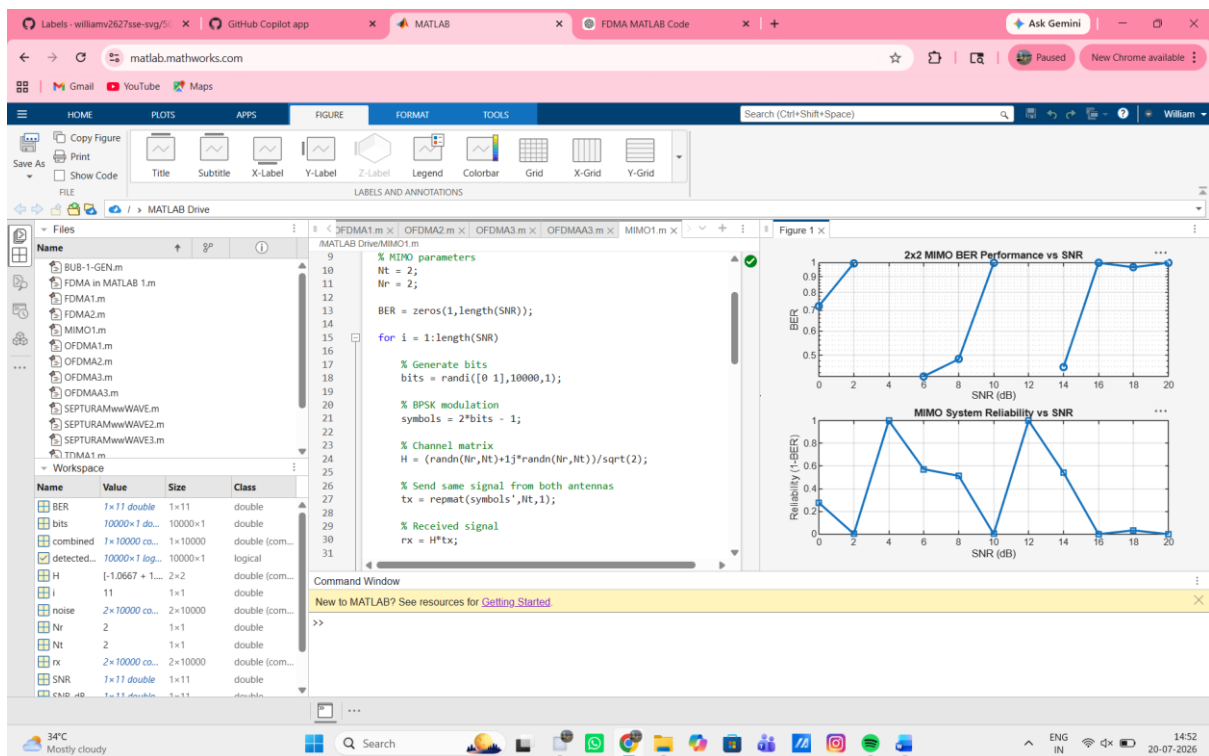
```
title('MIMO System Reliability vs SNR')
```

```
xlabel('SNR (dB)')
```

```
ylabel('Reliability (1-BER)')
```

```
grid on
```

OUTPUT:



```

(B)
clc;
clear;
close all;
% SNR range (dB)
SNR_dB = 0:2:20;
SNR = 10.^(SNR_dB/10);
% MIMO parameters
Nt = 2;
Nr = 2;
% Reliability metric
Reliability = zeros(1,length(SNR));
NoisePower = zeros(1,length(SNR));
for i = 1:length(SNR)
    % Simple Rayleigh channel model
    H = (randn(Nr,Nt) + 1j*randn(Nr,Nt))/sqrt(2);
    % Noise power
    NoisePower(i) = 1/SNR(i);
    % Reliability increases with SNR
    Reliability(i) = 1 - exp(-SNR(i)/10);
end
figure;
% ----- Graph 1: SNR vs Reliability -----
subplot(2,1,1);
plot(SNR_dB, Reliability, '-o', 'LineWidth', 2);
title('MIMO Reliability vs SNR');
xlabel('SNR (dB)');
ylabel('Reliability (0 to 1)');
grid on;
% ----- Graph 2: SNR vs Noise Effect -----
subplot(2,1,2);

```

```
plot(SNR_dB, NoisePower, '-s', 'LineWidth', 2);
```

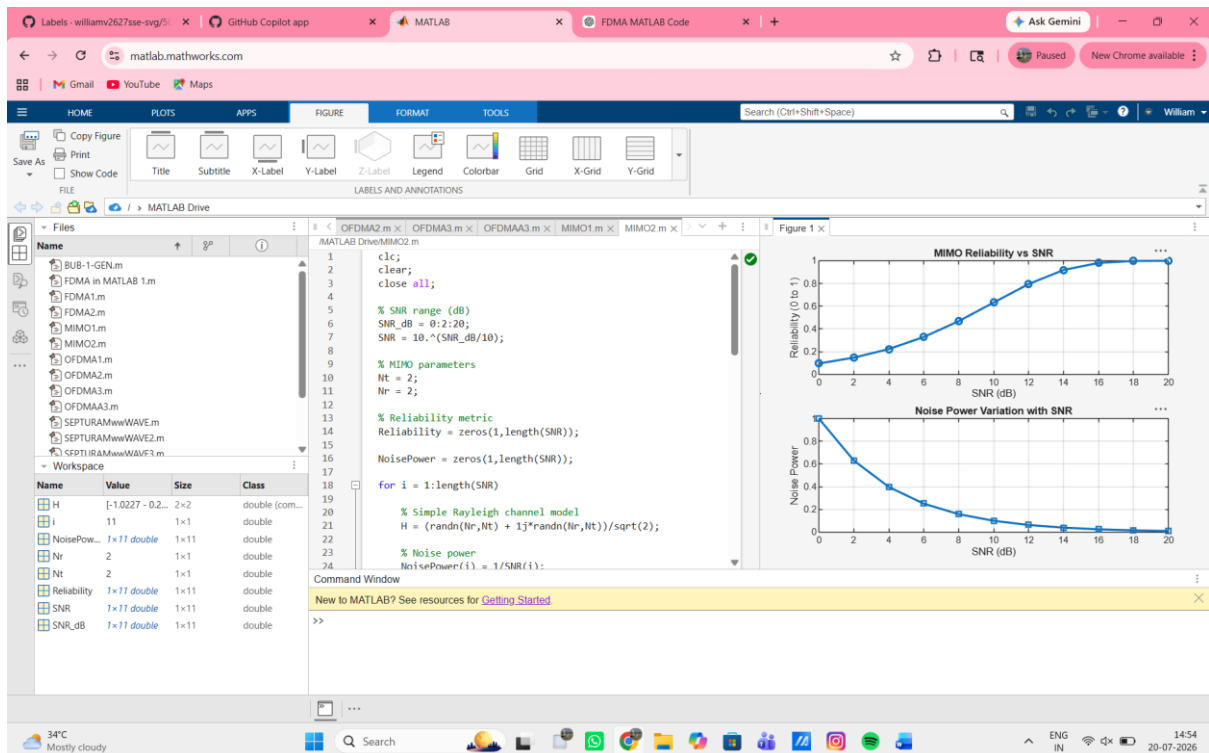
```
title('Noise Power Variation with SNR');
```

```
xlabel('SNR (dB)');
```

```
ylabel('Noise Power');
```

```
grid on;
```

OUTPUT:



(C)

```
clc;
```

```
clear;
```

```
close all;
```

```
% Resource Block parameters
```

```
rb_bw = 180e3; % 180 kHz per RB
```

```
% Users
```

```
users = 1:5;
```

```
% Allocated Resource Blocks per user
```

```
alloc_RB = [10 12 8 15 5];
```

```
% Modulation efficiency (bits/symbol)
```

```
% QPSK = 2, 16QAM = 4, 64QAM = 6
```

```
mod_eff = [2 4 6 4 2];
```

```

% Throughput calculation (Mbps)
throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;
% Display results
disp('User Throughput (Mbps):');
disp(throughput);
figure;
% ----- Graph 1: Throughput per User -----
subplot(2,1,1);
plot(users, throughput, '-o', 'LineWidth', 2);
title('OFDMA User Throughput');
xlabel('User Index');
ylabel('Throughput (Mbps)');
grid on;
% ----- Graph 2: RB vs Throughput -----
subplot(2,1,2);
plot(alloc_RB, throughput, '-s', 'LineWidth', 2);
title('Resource Blocks vs Throughput');
xlabel('Allocated Resource Blocks');
ylabel('Throughput (Mbps)');
grid on;
OUTPUT:

```

